E-commerce Product Analysis

- Analyzing online retail data to understand sales and customer behavior

Problem Definition

- Goal: Analyze transactions and predict total sales using machine learning

Dataset

- Online Retail Dataset
- Contains transactions, products, customers, and prices

Data Cleaning

- Handled missing values
- Removed duplicates
- Removed invalid data

EDA

- Top products
- Top countries
- Top customers
- Sales patterns

Feature Engineering

- Created TotalPrice
- Extracted Year, Month, Day, Hour
- Customer features added

Preprocessing

- Converted categorical data
- Scaled features
- Prepared dataset for ML

Model

- Used Random Forest Regressor
- Improved performance over Linear Regression

Evaluation

- Measured using R2 Score, MAE, MSE
- Achieved improved accuracy

Conclusion

- Data cleaning and feature engineering improved model performance
- Project demonstrates real-world data analysis